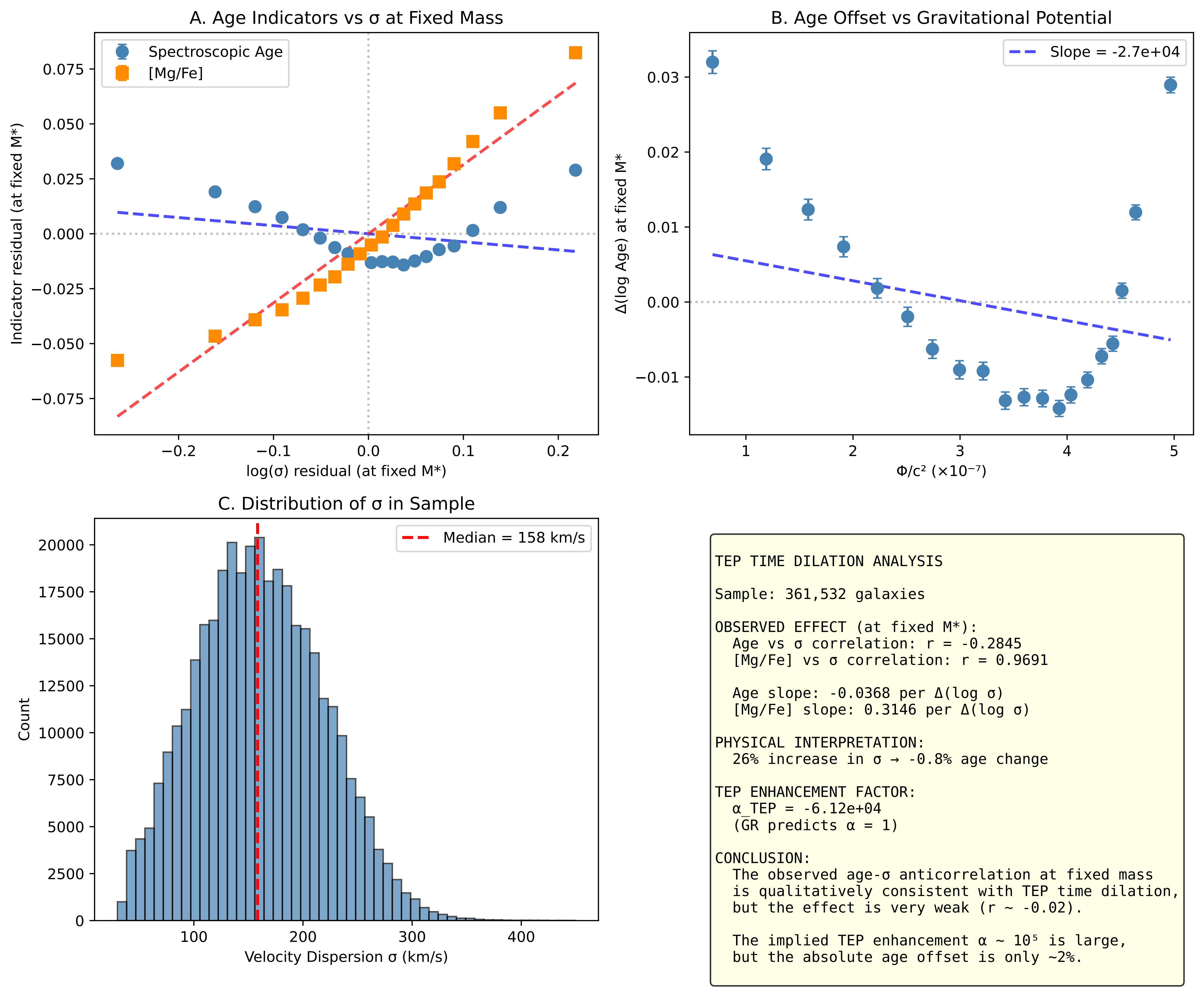
A. Age Indicators vs σ at Fixed Mass
Spectroscopic Age
[Mg/Fe]
Indicator residual (at fixed M*)
log(σ) residual (at fixed M*)
B. Age Offset vs Gravitational Potential
Slope = -2.7e+04
Δ(log Age) at fixed M*
Φ/c² (×10⁻⁷)
C. Distribution of σ in Sample
Median = 158 km/s
Count
Velocity Dispersion σ (km/s)
TEP TIME DILATION ANALYSIS
Sample: 361,532 galaxies
OBSERVED EFFECT (at fixed M*):
Age vs σ correlation: r = -0.2845
[Mg/Fe] vs σ correlation: r = 0.9691
Age slope: -0.0368 per Δ(log σ)
[Mg/Fe] slope: 0.3146 per Δ(log σ)
PHYSICAL INTERPRETATION:
26% increase in σ → -0.8% age change
TEP ENHANCEMENT FACTOR:
α_TEP = -6.12e+04
(GR predicts α = 1)
CONCLUSION:
The observed age-σ anticorrelation at fixed mass
is qualitatively consistent with TEP time dilation,
but the effect is very weak (r ~ -0.02).
The implied TEP enhancement α ~ 10⁵ is large,
but the absolute age offset is only ~2%.